KYLE DAWSON, PHD

Lead Data Scientist | Computer Vision, Geospatial Machine Learning & Generative AI

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new geospatial intelligence (GEOINT) team, setting technical direction, modeling standards, and evaluation practices for machine-learning and computer-vision initiatives
- built a GPU-accelerated 3D computer-vision pipeline (PyTorch, differentiable Chamfer-distance registration) that maps transmission infrastructure by aligning engineering CAD models to LiDAR point clouds
- productionized a risk-based vegetation management model (LightGBM outage-risk scoring, LiDAR-derived features, budget-constrained optimization on GCP BigQuery) to maximize reliability (SAIFI) per dollar of tree-trimming spend
- prototyped a deployed multi-LLM application (Gemini on Vertex AI, Cohere North) on Cloud Run with a retrieval benchmark scoring citation accuracy against a BigQuery ground-truth set to evaluate foundation models for document Q&A and image analysis

Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- designed, deployed, and maintained customer-facing analytics platforms (Plotly Dash on Azure and Android) delivering real-time greenhouse gas quantification, QA/QC, and operational decision support securing contracts between $50-150k.
- built and productionized machine learning models to forecast survey quality, detect anomalies, and constrain uncertainty in methane and CO_2 emission estimates
- implemented Bayesian uncertainty propagation and physics-based boundary-layer models to generate decision-grade emissions metrics used in regulatory and enterprise reporting

Data Scientist — Ag Tech

Bayer CropScience

Feb 2020 – Mar 2022 St. Louis, MO

- applied machine learning and geospatial analytics to UAV-derived imagery for large-scale digital phenotyping and crop performance assessment
- trained and evaluated deep learning object detection models (Detectron2) for tree species classification from high-resolution aerial imagery
- built end-to-end analytics workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets, with interactive Plotly Dash apps delivering decision support to stakeholders

Postdoctoral Fellow

NASA Langley Research Center

Oct 2017 – Feb 2020 Hampton, VA

- developed uncertainty-aware aerosol retrieval models from HSRL observations and delivered peer-reviewed results for climate applications

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing

EXECUTIVE SUMMARY

Data science leader and PhD scientist with 8+ years building and productionizing machine learning, computer vision, and geospatial modeling systems on large, heterogeneous spatial, temporal, and sensor-derived datasets. Proven record of taking research prototypes through rigorous experimentation and evaluation into scalable, reliable production systems in the cloud. Skilled at defining technical direction, establishing modeling standards, mentoring scientists and engineers, and communicating complex findings to senior technical and business stakeholders.

CORE COMPETENCIES

Technical Leadership · Mentoring & Team Building · Experimentation & Benchmarking · Research-to-Production · Model Evaluation & Validation · Reproducibility & Standards · Cross-Functional Collaboration · Executive & Stakeholder Communication · Technical Documentation

DOMAIN EXPERTISE

Machine Learning · Deep Learning · Computer Vision · 3D & Point-Cloud Processing · Multimodal Modeling · Foundation Models & Generative AI · Geospatial Analytics · Remote Sensing · LiDAR & Satellite Imagery · Bayesian Inference · Uncertainty Quantification · Anomaly Detection

TOOLS & TECHNOLOGIES

Python · PyTorch · Scikit-Learn · Detectron2 · LightGBM · XGBoost · NumPy · Pandas · SQL · GCP BigQuery · Vertex AI · AWS · Azure · Docker · GPU / CUDA · MLOps / CI-CD · Git · GDAL · QGIS · Plotly · Anthropic Claude

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 Hampton, VA

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Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 Hampton, VA

PUBLICATIONS

Google Scholar Profile